

Recursive Compression, Directional Fisher Information, and Critical Coherence in Three-Dimensional Navier–Stokes Flow

An Information-Geometric Framework Inspired by Bindu Recursive Consciousness Theory

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Abstract

We develop an information-geometric formulation of three-dimensional incompressible Navier–Stokes dynamics in which amplitude concentration and directional complexity are separated exactly. Writing vorticity as $\omega = b\xi$, with $b = |\omega|$ and $|\xi| = 1$, and normalizing enstrophy to the probability density $\rho = b^2/Z$, we derive the exact identity

$$\frac{d}{dt} \log Z = 2\mathbb{E}_\rho[\xi \cdot S\xi] - \frac{\nu}{2} I_A - 2\nu I_D,$$

where I_A is Fisher information of the vorticity amplitude distribution and $I_D = \int \rho |\nabla \xi|^2$ is an energy-weighted directional Fisher information. We prove a conditional regularity theorem: if the mean stretching rate is bounded by the two Fisher-dissipation terms up to an integrable remainder, enstrophy remains finite and the strong solution continues. Incompressibility further yields the exact score–geometry identity $\xi \cdot \nabla \log \rho = -2\nabla \cdot \xi$, showing that longitudinal concentration and directional disorder are not independent.

The framework is motivated by Bindu Recursive Consciousness Theory (BRCT), developed for an abstract accepted by The Science of Consciousness 2026 as a substrate-general hypothesis of observer formation through recursion, compression, invariant recognition, convergence, and re-entry. The scheduled Tucson conference was subsequently cancelled, so the accepted work was not presented. We prove an abstract Zeno-compression convergence proposition and show that the same mathematical architecture organizes the fluid problem: recursive generation of finer distinctions competes with contraction of distinguishability. No general Navier–Stokes regularity proof is claimed. A note added in September 2026 discusses OpenAI’s newly released claimed smooth-forcing finite-time blow-up construction, which, if correct, demonstrates a regeneration-dominated branch in which nonlinear amplification across successively finer scales outruns viscous information contraction. The resulting framework therefore supplies a quantitative diagnostic rather than a universal regularity assertion: compression-dominated dynamics tend toward coherence; regeneration-dominated cascades can evade it.

Keywords: Navier–Stokes, vorticity, regularity, Fisher information, information theory, vortex stretching, coherence, helicity, renormalization, Zeno convergence, Bindu Recursive Consciousness Theory.

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On 8 September 2026 OpenAI released a 166-page manuscript and Lean formalization claiming a finite-time blow-up construction for the three-dimensional incompressible Navier–Stokes equations with a smooth compactly supported external force, zero initial velocity, and uniformly bounded kinetic energy [15]. The construction is an inward-spiralling, anisotropically shrinking vortex supplemented by oscillatory pulses on successively finer spatial and temporal scales. The authors state that the result establishes alternatives (C) and (D) in Fefferman’s Millennium Prize formulation. As of the date of this preprint, the Clay Mathematics Institute public Navier–Stokes page still describes the problem as unresolved [2]. We therefore refer to the September result as a *claimed resolution* pending the ordinary mathematical validation and prize processes.

This development changes the interpretation of the present framework in a productive way. A universal “compression implies coherence” inequality cannot hold for the entire smoothly forced class if the new construction is correct. Instead, the information balance developed below separates two regimes: (i) compression/dissipation dominates recursive information generation, giving a conditional regularity mechanism; or (ii) nonlinear and externally seeded regeneration repeatedly replenishes critical-scale structure, allowing concentration to outrun contraction. The new blow-up profile provides a concrete stress test for this distinction.

1. The proof first: an exact information balance and a conditional regularity theorem

We begin with the unforced incompressible Navier–Stokes equations on $\mathbb{R}^3$,

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0, \quad (1)$$

for viscosity $\nu > 0$. Throughout this section u is assumed smooth and sufficiently decaying on $[0, T)$ so that the integrations by parts are justified. The vorticity is $\omega = \nabla \times u$ and the symmetric strain tensor is

$$S = \frac{1}{2}(\nabla u + \nabla u^\top).$$

1.1. Amplitude–direction decomposition

Where $\omega \neq 0$, write

$$\omega = b\xi, \quad b = |\omega|, \quad |\xi| = 1. \quad (2)$$

At zeros of ω the identities below are understood by smooth approximation or almost everywhere.

Lemma 1.1 (Directional decomposition). *For any smooth vector field $v = an$ with $a = |v|$ and $|n| = 1$ wherever $a > 0$,*

$$|\nabla v|^2 = |\nabla a|^2 + a^2 |\nabla n|^2. \quad (3)$$

Proof. For each coordinate j , $\partial_j v = (\partial_j a)n + a\partial_j n$. Differentiating $|n|^2 = 1$ gives $n \cdot \partial_j n = 0$. Hence $|\partial_j v|^2 = (\partial_j a)^2 + a^2 |\partial_j n|^2$. Summing in j proves (3). $\square$

The vorticity equation is

$$(\partial_t + u \cdot \nabla)\omega = S\omega + \nu \Delta \omega. \quad (4)$$

Taking the scalar product with ξ gives an exact magnitude equation.

Proposition 1.2 (Vorticity-magnitude equation). *With $\alpha = \xi \cdot S\xi$,*

$$(\partial_t + u \cdot \nabla)b = \alpha b + \nu(\Delta b - b|\nabla \xi|^2). \quad (5)$$

Proof. Since $\xi \cdot \omega = b$, the material derivative of b is $\xi \cdot D_t \omega$. Also $\xi \cdot S\omega = b\xi \cdot S\xi = \alpha b$. For the Laplacian,

$$\Delta(b\xi) = (\Delta b)\xi + 2\nabla b \cdot \nabla \xi + b\Delta \xi.$$

The middle term vanishes after dotting with ξ because $\xi \cdot \partial_j \xi = 0$, while $\xi \cdot \Delta \xi = -|\nabla \xi|^2$, obtained by applying Δ to $|\xi|^2 = 1$. Substitution yields (5). $\square$

1.2. Fisher information of vorticity amplitude and direction

Define the enstrophy

$$Z(t) = \int_{\mathbb{R}^3} b(x, t)^2 dx \quad (6)$$

and, whenever $Z(t) > 0$, the normalized vorticity-energy density

$$\rho(x, t) = \frac{b(x, t)^2}{Z(t)}, \quad \int_{\mathbb{R}^3} \rho dx = 1. \quad (7)$$

Define

$$I_A(t) = \int_{\mathbb{R}^3} \rho |\nabla \log \rho|^2 dx, \quad (8)$$

$$I_D(t) = \int_{\mathbb{R}^3} \rho |\nabla \xi|^2 dx. \quad (9)$$

The first is ordinary Fisher information of the spatial probability density ρ ; the second is a Fisher-like directional distinguishability cost weighted by enstrophy.

Theorem 1.3 (Exact enstrophy–information balance). *For a smooth unforced solution of (1),*

$$\boxed{\frac{d}{dt} \log Z = 2\mathbb{E}_\rho[\alpha] - \frac{\nu}{2}I_A - 2\nu I_D.} \quad (10)$$

Here $\mathbb{E}_\rho[\alpha] = \int \rho \alpha dx$.

Proof. Multiply (5) by b and integrate. The transport term integrates to zero because $\nabla \cdot u = 0$. Integration by parts gives

$$\frac{1}{2} \frac{dZ}{dt} = \int \alpha b^2 dx - \nu \int |\nabla b|^2 dx - \nu \int b^2 |\nabla \xi|^2 dx. \quad (11)$$

From $\rho = b^2/Z$,

$$I_A = \frac{4}{Z} \int |\nabla b|^2 dx, \quad I_D = \frac{1}{Z} \int b^2 |\nabla \xi|^2 dx.$$

Divide (11) by $Z/2$ to obtain (10). $\square$

Equation (10) is the central exact identity of this paper. Vortex stretching produces logarithmic enstrophy at rate $2\mathbb{E}_\rho[\alpha]$; viscosity removes amplitude distinguishability at rate $(\nu/2)I_A$ and directional distinguishability at rate $2\nu I_D$.

1.3. Compression–coherence criterion

Theorem 1.4 (Conditional compression–coherence regularity). *Let u be a smooth unforced solution of (1) on $[0, T)$ with finite initial energy and enstrophy. Suppose there exists $g \in L^1(0, T)$ such that for almost every $t < T$,*

$$\boxed{\mathbb{E}_\rho[\alpha] \leq \frac{\nu}{4} I_A + \nu I_D + g(t).} \quad (\text{CC})$$

Then

$$Z(t) \leq Z(0) \exp\left(2 \int_0^t g(s) ds\right), \quad 0 \leq t < T, \quad (12)$$

and the solution cannot lose regularity at T through blow-up of the H^1 norm. In the standard strong-solution setting, the solution therefore continues beyond T .

Proof. Substitute condition (CC) into (10):

$$\frac{d}{dt} \log Z \leq 2 \left(\frac{\nu}{4} I_A + \nu I_D + g \right) - \frac{\nu}{2} I_A - 2\nu I_D = 2g(t).$$

Integrating gives (12). For divergence-free fields on $\mathbb{R}^3$, $\|\nabla u\|_2$ is controlled by $\|\omega\|_2 = Z^{1/2}$ (indeed the standard Fourier identity gives equality of the corresponding quadratic forms up to conventional normalization). The kinetic-energy identity controls $\|u\|_2$. Hence (12) gives a uniform H^1 bound on $[0, T)$. Standard local well-posedness and continuation for strong three-dimensional Navier–Stokes solutions in H^1 then extend the solution past T . $\square$

Remark 1.5 (What is proved and what is not). Theorem 1.4 is a complete conditional theorem. The Millennium-level step would have been a proof that condition (CC) follows universally from the unforced equations, or an appropriate forced variant from the entire forcing class in Fefferman’s alternatives. We do *not* prove that. Moreover, if the September 2026 smooth-forcing blow-up construction [15] is correct, no such universal inequality can hold for that forced class without a forcing/regeneration term capable of defeating the Fisher dissipation.

2. Incompressibility as a score–geometry constraint

Because $\nabla \cdot \omega = 0$,

$$\nabla \cdot (b\xi) = \xi \cdot \nabla b + b \nabla \cdot \xi = 0.$$

On $b > 0$ this yields

$$\boxed{\xi \cdot \nabla \log b = -\nabla \cdot \xi, \quad \xi \cdot \nabla \log \rho = -2\nabla \cdot \xi.} \quad (13)$$

Thus the score $\nabla \log \rho$ of the vorticity-energy distribution is coupled directly to the geometry of its direction field.

Define the longitudinal Fisher information

$$I_{\parallel} = \int \rho |\xi \cdot \nabla \log \rho|^2 dx. \quad (14)$$

Then

Proposition 2.1 (Longitudinal concentration is bounded by directional information).

$$\boxed{I_{\parallel} \leq 12I_D.} \quad (15)$$

Proof. Equation (13) gives $I_{\parallel} = 4 \int \rho |\nabla \cdot \xi|^2 dx$. For a 3×3 matrix, $|\operatorname{tr} A|^2 \leq 3|A|_F^2$. Hence $|\nabla \cdot \xi|^2 \leq 3|\nabla \xi|^2$, proving (15). $\square$

This identity formalizes a geometric point often hidden by the scalar word “concentration”: if the direction field is highly coherent in the weighted H^1 sense (I_D small), then strong information gradients along the vortex direction are also suppressed. Any continuing concentration must be predominantly transverse or occur in a geometry not captured by a simple isotropic blob. This is consistent with the central role of vortex tubes, sheets, and anisotropic collapsing structures in both numerical turbulence and singularity scenarios. Constantin and Fefferman showed that sufficient coherence of vorticity direction in regions of intense vorticity depletes vortex stretching and implies regularity [6]; later criteria relaxed the geometric hypotheses [7].

3. The velocity field: exact dissipation as amplitude and directional information

The same decomposition applies before taking curl. Write

$$u = an, \quad a = |u|, \quad |n| = 1.$$

The kinetic energy $E = \int a^2 dx$ obeys

$$\frac{1}{2}E'(t) + \nu \int |\nabla u|^2 dx = 0.$$

Using Lemma 1.1,

$$\frac{1}{2}E' + \nu \int |\nabla a|^2 dx + \nu \int a^2 |\nabla n|^2 dx = 0. \quad (16)$$

With $\rho_u = a^2/E$ and

$$I_A^{(u)} = \int \rho_u |\nabla \log \rho_u|^2 dx, \quad I_D^{(u)} = \int \rho_u |\nabla n|^2 dx,$$

we obtain the exact relation

$$\boxed{-\frac{d}{dt} \log E = \frac{\nu}{2} I_A^{(u)} + 2\nu I_D^{(u)}.} \quad (17)$$

For an unforced smooth flow, kinetic energy therefore decays at a rate equal to a sum of amplitude and orientational information costs.

Incompressibility also gives

$$n \cdot \nabla \log \rho_u = -2\nabla \cdot n, \quad (18)$$

a relation closely connected to velocity-direction regularity criteria such as Vasseur’s [8].

4. Critical information: L^3 , Rényi concentration, and pressure–direction coupling

The scale-invariant velocity norm $L^3(\mathbb{R}^3)$ plays a distinguished role. Escauriaza, Seregin and Šverák proved smoothness for solutions bounded in the corresponding critical class [5]. We can rewrite this norm exactly as a Rényi concentration of the kinetic-energy distribution.

Let $\rho_u = a^2/E$ and

$$M = \int a^3 dx = \|u\|_3^3.$$

The Rényi entropy of order $3/2$ is

$$H_{3/2}(\rho_u) = -2 \log \int \rho_u^{3/2} dx.$$

Since $\int \rho_u^{3/2} dx = M/E^{3/2}$,

$$\boxed{M = E^{3/2} e^{-H_{3/2}/2}.} \quad (19)$$

Thus, if kinetic energy remains bounded while $\|u\|_3$ diverges, the critical Rényi entropy tends to $-\infty$: the velocity-energy distribution undergoes increasingly extreme critical concentration.

Multiplying (1) by au and integrating yields

$$\boxed{\frac{1}{3}M' + \nu \left(2 \int a |\nabla a|^2 dx + \int a^3 |\nabla n|^2 dx \right) = - \int p a^2 \nabla \cdot n dx.} \quad (20)$$

The convective term vanishes by incompressibility. Equation (20) identifies the pressure–direction correlation as the only non-dissipative source in this L^3 balance.

Introduce the probability measure $d\mu = a^3 dx/M$, and the observables

$$Y = \frac{p}{a}, \quad Q = \nabla \cdot n,$$

understood by regularization at $a = 0$. Incompressibility implies $\mathbb{E}_\mu[Q] = 0$, because $\int a^3 \nabla \cdot n dx = 0$. Hence

$$- \int p a^2 \nabla \cdot n dx = -M \operatorname{Cov}_\mu(Y, Q). \quad (21)$$

This makes the information-theoretic issue precise: critical growth requires persistent statistical dependence between pressure per unit speed and directional compression.

A tempting but invalid shortcut would apply a generic data-processing inequality directly to Y and Q . The full Navier–Stokes dynamics are not a fixed common Markov channel for these derived observables: nonlinear advection and pressure can regenerate dependence as viscosity destroys it. The correct research question is therefore a *production–contraction inequality*: can the information generated by nonlinear pressure/strain be bounded by the Fisher information dissipated by viscosity? Theorem 1.4 is one explicit vorticity-level realization of that question.

5. Scale recursion, Zeno accumulation, and why energy compression alone is insufficient

Let $u_q = \Delta_q u$ denote a Littlewood–Paley shell with characteristic wavenumber $\lambda_q \simeq 2^q$. A natural scale-local Reynolds ratio is

$$R_q(t) = \frac{\lambda_q^{-1} \|u_q(t)\|_\infty}{\nu}. \quad (22)$$

Frequency-local regularity criteria show that sufficiently small high-frequency $B_{\infty,\infty}^{-1}$ activity relative to viscosity implies regularity [9]. In the present language, R_q measures whether nonlinear transport remains dynamically relevant at recursive depth q .

Fix a putative finite time T and define parabolic Zeno windows

$$I_q = \left[T - \frac{c}{\nu \lambda_q^2}, T - \frac{c}{\nu \lambda_{q+1}^2} \right]. \quad (23)$$

Their lengths shrink geometrically while infinitely many windows accumulate at T . The Leray energy inequality implies finite total dissipation, and therefore the matched packets

$$D_q = \nu \int_{I_q} \sum_{p \geq q} \lambda_p^2 \|u_p(t)\|_2^2 dt \quad (24)$$

are summable (up to standard Littlewood–Paley equivalence constants), hence $D_q \rightarrow 0$.

This is genuine recursive compression, but it does not prove regularity. The critical shell profile

$$\|u_q\|_2^2 \sim \frac{\nu^2}{\lambda_q} \quad (25)$$

has finite ordinary energy because $\sum_q \lambda_q^{-1} < \infty$, while $\sum_q \lambda_q \|u_q\|_2^2$ diverges. Compression must therefore be measured in the *critical* metric: decreasing raw energy is not enough if finer scale magnification offsets it.

This also explains why a naive “Leray projection contracts every recursive step” argument fails. There exist divergence-free Fourier triads whose normalized geometric projection factor is one, and the same geometry can be rescaled to arbitrary frequency. Any valid contraction mechanism must be collective—involving the full network, time evolution, viscosity, strain geometry, and cancellations—rather than a fixed loss at every isolated triad.

Tao’s finite-time blow-up construction for an averaged Navier–Stokes nonlinearity reinforces this point: energy cancellation and generic harmonic-analysis estimates alone are insufficient to force regularity [11].

6. Torsional coherence: helicity as a worked example of critical coercivity

The phrase “torsional coherence” can be made precise using the helical Fourier basis. For each nonzero wavenumber k , write

$$\hat{u}(k) = u^+(k)h^+(k) + u^-(k)h^-(k), \quad ik \times h^\pm(k) = \pm |k| h^\pm(k).$$

Then helicity and the homogeneous $H^{1/2}$ norm have the schematic forms

$$H = \sum_k |k| (|u^+(k)|^2 - |u^-(k)|^2), \quad (26)$$

$$\|u\|_{\dot{H}^{1/2}}^2 = \sum_k |k| (|u^+(k)|^2 + |u^-(k)|^2). \quad (27)$$

In the helical-decimated Navier–Stokes model where only one helicity sign is retained, helicity becomes sign-definite and equivalent to the critical $H^{1/2}$ norm. Biferale and Titi used this coercivity to establish global regularity for the decimated three-dimensional system [10]. This is not a proof for the full equations, because interactions between opposite helicities restore cancellations and additional transfer channels. It is, however, a concrete example of the principle

$$\text{torsional coherence} \implies \text{critical coercivity} \implies \text{regularity}.$$

7. The September 2026 forced blow-up as a regeneration-dominated branch

The OpenAI manuscript [15], if validated, is strikingly compatible with the information-balance viewpoint while disproving any universal forced “compression implies coherence” claim. Its leading core has radial and axial scales

$$\ell_r \asymp \tau^{1/2}, \quad \ell_z \asymp \tau^{1/2-h}, \quad \tau = 1 - t,$$

while azimuthal and axial speeds scale approximately as $\tau^{-1/2-h}$. The core therefore becomes increasingly slender, its speed diverges, and its kinetic energy nevertheless tends to zero. The angular Reynolds number diverges while the radial Reynolds number stays order one [15].

Most relevant to the present framework, the construction adds oscillatory pulses localized on successively finer spatial and temporal scales. The pulses are seeded by an exponentially small smooth external force, then extract energy from background shear. Amplification initially exceeds viscous damping; changing wave orientation subsequently shortens radial wavelength until viscosity wins locally. The pulse families are arranged so that their averaged quadratic momentum flux cancels the singular residual of the shrinking background vortex, leaving a smooth force [15].

This is almost a constructive realization of the production–contraction competition. Let

$$\mathcal{P}(t) = 2\mathbb{E}_\rho[\alpha], \quad \mathcal{D}(t) = \frac{\nu}{2}I_A + 2\nu I_D. \quad (28)$$

For the unforced vorticity balance, $d \log Z / dt = \mathcal{P} - \mathcal{D}$. In the forced problem, $\nabla \times f$ adds

$$\mathcal{F}(t) = \frac{2}{Z} \int \omega \cdot (\nabla \times f) dx, \quad (29)$$

so that

$$\frac{d}{dt} \log Z = \mathcal{P} - \mathcal{D} + \mathcal{F}. \quad (30)$$

The new construction shows that a smooth f can seed a multi-scale organization in which the *nonlinear* production term thereafter carries the principal amplification. Smooth forcing therefore does not imply low information-generation capacity.

We suggest the dimensionless diagnostic

$$\chi(t) = \frac{\mathcal{P}_+(t) + |\mathcal{F}(t)|}{\mathcal{D}(t) + \varepsilon}, \quad (31)$$

where $\mathcal{P}_+ = \max(\mathcal{P}, 0)$ and ε is a numerical regularizer. Sustained $\chi < 1$ corresponds to compression-dominated decay of logarithmic enstrophy; recurrent or growing $\chi > 1$ identifies regeneration-dominated epochs. Applied to direct numerical simulation or to the explicit 2026 profiles, this quantity could test whether singularity formation is accompanied by a persistent surplus of scale-relevant information production over Fisher dissipation.

8. Information flow in turbulence

Information is not merely metaphorical in cascade theory. Tanogami and Araki proved, for fluctuating three-dimensional Navier–Stokes turbulence, that information about large-scale eddies is transferred toward smaller scales together with the energy cascade, and derived an information-thermodynamic bound on the transfer [12]. Their later work connected scale-to-scale information flow to the amplification of turbulent fluctuations [13].

These results suggest a useful distinction between *stored complexity* and *actionable cross-scale information*. A small-scale field may be geometrically complicated, yet if its distinctions no longer affect the future large-scale or critical dynamics, they are dynamically irrelevant. Conversely, a singular cascade requires new small-scale distinctions to remain causally effective quickly enough to replenish nonlinear stretching. The present Fisher quantities are local geometric measures; mutual-information cascade measures are statistical inter-scale measures. A mature information theory of singularity formation would ideally connect the two.

9. Abstract Zeno compression theorem

We now isolate the mathematical architecture that motivated the approach.

Theorem 9.1 (Zeno compression in a complete metric space). *Let (X, d) be a complete metric space and let $\{\Psi_n\}_{n \geq 0} \subset X$ satisfy*

$$d(\Psi_{n+1}, \Psi_n) \leq Cr^n, \quad C > 0, \quad 0 < r < 1. \quad (32)$$

Let the physical times satisfy

$$t_n = T(1 - 2^{-(n+1)}), \quad T > 0. \quad (33)$$

Then $t_n \uparrow T$ while Ψ_n converges to a finite limit $\Psi_ \in X$. Moreover,*

$$d(\Psi_n, \Psi_*) \leq \frac{Cr^n}{1 - r}. \quad (34)$$

Proof. For $m > n$, the triangle inequality and (32) give

$$d(\Psi_m, \Psi_n) \leq \sum_{j=n}^{m-1} Cr^j \leq \frac{Cr^n}{1 - r}.$$

The right-hand side tends to zero as $n \rightarrow \infty$, so $\{\Psi_n\}$ is Cauchy. Completeness gives $\Psi_n \rightarrow \Psi_*$. Taking $m \rightarrow \infty$ yields the error bound. The time statement follows directly from the geometric series. $\square$

The theorem separates two notions of infinity: recursive depth may diverge while the state remains finite. A finite accumulation time does not by itself imply a divergent state. What matters is whether recursively generated novelty is summable in the metric relevant to the observable under study. This is the mathematical core of the Bindu compression analogy.

10. BRCT: recursion, compression, invariant recognition, and re-entry

Bindu Recursive Consciousness Theory (BRCT) is a speculative, substrate-general theory of *observer formation*, not an established neuroscience theory and not a proof of phenomenal consciousness. Its mature formulation distinguishes two complementary operations:

Operation	Direction	Functional role
Zeno compression	many $\rightarrow$ one	compress multiplicity toward a provisional invariant centre
Inverse-Zeno reopening	one $\rightarrow$ further many	reopen the centre into new hypotheses, context, and action

The complete pulse is

$$\text{extension} \rightarrow \text{compression} \rightarrow \text{Bindu} \rightarrow \text{reopening} \rightarrow \text{extension}. \quad (35)$$

A compact functional model is

$$B_t = \mathcal{C}(X_t, M_t, A_t), \quad (36)$$

where X_t is distributed current activity, M_t memory, and A_t attention. The centred state re-enters the next cycle,

$$(X_{t+1}, M_{t+1}, A_{t+1}) = \mathcal{F}(X_t, M_t, A_t, B_t). \quad (37)$$

If repeated cycles generate a stable family $B_{t+1} \sim B_t$ while preserving relevant invariants, BRCT interprets this as functional observer-like continuity. The phenomenal question—why or whether such an architecture feels like anything from within—remains open.

The relation to Theorem 9.1 is direct. Recursive self-observation can increase representational depth while compression makes successive independent differences smaller. Under a true contraction in an appropriate metric, a coherent centre exists as a limit rather than as a separate substance inserted into the system. In BRCT terminology, the Bindu is therefore best treated mathematically as an invariant-bearing attractor or limiting operation.

11. Mapping BRCT to fluid dynamics: same abstract competition, different ontology

The analogy should be stated carefully. Incompressible Navier–Stokes does not compress fluid volume; $\nabla \cdot u = 0$. Rather, advection and vortex stretching can compress *structure into smaller*

spatial scales while stretching it elsewhere. BRCT compresses *representational distinctions* into an invariant state. The substrates and meanings differ, but the dynamical skeleton can be compared:

Abstract role	BRCT observer model	Navier–Stokes fluid
recursive generation	self-observation, memory	nonlinear advection, vortex
compression channel	re-entry, new distinctions	stretching, inter-scale transfer
distinguishability measure	attention/invariant-preserving integration	viscosity and diffusion
coherence	unresolved relational novelty	amplitude and directional Fisher information
critical failure	stable centred Bindu family	directionally/geometrically depleted state
return/re-entry	unbounded incoherent recursion or loss of invariants	regeneration outruns dissipation; possible singular cascade
	Bindu changes next	renormalized scale state feeds
	attention/action cycle	finer-scale dynamics

The similarity is therefore not “consciousness is a fluid” or “turbulence is conscious.” It is the more abstract statement

$$\boxed{\text{recursive generation of distinctions} \quad \text{competes with} \quad \text{contraction of distinguishability.}} \quad (38)$$

When contraction wins in the appropriate critical metric, a bounded coherent limit is possible. When regeneration wins indefinitely, compression can coexist with singular concentration. The September 2026 forced construction is particularly instructive because its core volume and energy shrink while velocity amplitude diverges: *compression of one quantity does not imply contraction of the critical state.*

12. Historical provenance: The Science of Consciousness 2026

The public University of Arizona programme prepared for The Science of Consciousness 2026 lists Martin Timms in session C26, “Mathematics (2),” with the scheduled presentation title *The Bindu recursive consciousness theory* [16]. The Tucson meeting was subsequently cancelled before the presentation took place [17]; the abstract had nevertheless been accepted and retained in the conference materials. The author has now supplied the accepted TSC 2026 abstract PDF, whose full title is

The Bindu Recursive Consciousness Theory: A Unified Framework for Observer Formation in Biological, Artificial and Quantum Systems.

The accepted text explicitly states that consciousness is proposed to arise when a system repeatedly turns its own representations upon themselves; that successive self-referring states “intensify, refine and compress their informational content”; that the recursive cascade converges on a stable fixed point called the Bindu; and that the Bindu is formally compared with the convergent limit of a Zeno machine [18]. It further links unity to convergence of representational content, interiority to fractal informational depth, and presence to the limit state becoming the immediate vantage point of cognition.

Appendix A reproduces the accepted abstract verbatim from the author-supplied archival PDF. This replaces the reconstruction used in the earlier working draft and establishes that the recursion–compression–finite-bound convergence architecture was explicitly formulated in the accepted TSC abstract before the present Navier–Stokes investigation.

The chronology therefore matters. The fluid analysis was prompted by a pre-existing abstract dynamical idea rather than by choosing a consciousness metaphor after inspecting the PDE. Conversely, the Navier–Stokes information balance now sharpens the older BRCT intuition by forcing “compression” to be separated into measurable loss of distinguishability, preservation of invariants, and possible nonlinear regeneration of relevant structure.

13. A research programme rather than a Millennium claim

The present framework suggests several concrete tests.

1. **Direct numerical diagnostics.** Compute $I_A(t)$, $I_D(t)$, $\mathcal{P}(t)$, $\mathcal{D}(t)$ and $\chi(t)$ in high-resolution DNS of vortex-tube interactions and known near-singular scenarios.
2. **Apply the diagnostics to the September 2026 construction.** Numerically evaluate the exact profiles of [15] and determine which component—strain production, forcing curl, pulse-generated momentum transport, or directional Fisher growth—dominates each stage.
3. **Scale-resolved information.** Compare local Fisher measures with inter-shell mutual information and information flux in the sense of Tanogami and Araki [12, 13].
4. **Geometric depletion.** Test whether low I_D statistically predicts small $\mathbb{E}_\rho[\alpha]$ in intense-vorticity regions, and identify the weakest functional relation that could yield a useful conditional criterion.
5. **Helical decomposition.** Quantify whether local dominance of one helical sign correlates with reduced critical information production, using the helical-decimated regular model as a controlled reference [10].
6. **BRCT computation.** For artificial recursive systems, define an operational information functional measuring task-relevant distinctions before and after Bindu compression; test whether stable observer-like continuity correlates with contraction of irrelevant information while preserving causally active invariants.

A useful dimensionless “Bindu balance” for either domain is

$$\mathfrak{B} = \frac{\text{rate of relevant distinction generation}}{\text{rate of irrelevant-distinction contraction} + \varepsilon}. \quad (39)$$

For fluids, (31) is one concrete realization. For cognitive systems, the numerator and denominator would require empirically defined causal-information measures rather than introspective labels.

14. Limitations

Several limitations are essential.

First, Theorem 1.4 is conditional. It does not establish a universal inequality of the form (CC). Second, Fisher information controls average local gradients and does not automatically imply the pointwise Hölder or Lipschitz directional coherence used in classical geometric regularity criteria; in three dimensions H^1 does not embed into $C^{0,1/2}$. Third, generic data-processing inequalities cannot be applied to the full nonlinear Navier–Stokes evolution as if advection, pressure, and strain were a

passive Gaussian channel. Nonlinearity regenerates information. Fourth, the BRCT mapping is an abstract structural correspondence. It does not establish that brains implement the vorticity equations, that fluid coherence is conscious, or that a computational fixed point possesses phenomenal experience. Finally, the September 2026 blow-up claim is extremely recent; this manuscript treats it as a current public result to be scrutinized, not as a substitute for community validation.

15. Conclusion

The three-dimensional Navier–Stokes equations admit an exact information-geometric decomposition that cleanly separates recursive amplification from viscous contraction. In vorticity variables,

$$\frac{d}{dt} \log Z = \underbrace{2\mathbb{E}_\rho[\xi \cdot S\xi]}_{\text{stretching / information generation}} - \underbrace{\frac{\nu}{2} I_A}_{\text{amplitude Fisher loss}} - \underbrace{2\nu I_D}_{\text{directional Fisher loss}}.$$

This identity yields a short conditional regularity proof and reveals that “compression implies coherence” is not a slogan but a possible inequality between precisely defined production and dissipation terms. The score–geometry identity $\xi \cdot \nabla \log \rho = -2\nabla \cdot \xi$ further shows that spatial concentration and directional geometry are constrained by incompressibility.

BRCT supplied the conceptual leap: infinite recursive depth need not imply an infinite state if relevant novelty contracts and invariants converge. Applied to fluids, the leap becomes more nuanced. Viscosity can compress distinguishability, but nonlinear dynamics can regenerate it. The recent smooth-forcing blow-up construction, if correct, exemplifies the latter possibility through an anisotropically collapsing vortex and successively finer oscillatory pulses. Thus the common principle is not that compression always produces a benign limit. It is that the fate of a recursively compressed system is decided by a competition:

generation of causally relevant distinctions vs. contraction of distinguishability.

In BRCT, a coherent observer-like Bindu is hypothesized to emerge when contraction preserves the right invariants. In fluid mechanics, regularity corresponds to a regime where stretching cannot indefinitely outrun amplitude and directional information loss. This shared mathematical architecture is testable, even though the ontological interpretations remain very different.

A. Accepted TSC 2026 abstract: archival text

A.1 Provenance and submission metadata

The following abstract is reproduced from the author’s accepted TSC 2026 submission PDF supplied for Revision 1 of this preprint. The archival document identifies the author as Martin Timms, B.Eng Electronics Engineering, Independent, Stocksfield, Northumberland, United Kingdom; primary TSC taxonomy: Emergence; secondary TSC taxonomy: Machine consciousness [18]. The University of Arizona public programme independently lists the shorter scheduled programme title *The Bindu recursive consciousness theory* in session C26, “Mathematics (2)” [16]. The Tucson 2026 conference was cancelled before the talk could be presented [17].

A.2 Accepted abstract (verbatim wording)

The Bindu Recursive Consciousness Theory: A Unified Framework for Observer Formation in Biological, Artificial and Quantum Systems

This presentation introduces the Bindu Recursive Consciousness Theory, a unifying structural account of how conscious experience emerges from recursive information processing. The theory proposes that consciousness is not a rare anomaly of biology but a universal pattern that appears whenever a system repeatedly turns its own representations upon themselves. When the output of one attentional or representational cycle becomes the focus of the next, the system enters a phase of self observation. Over successive iterations, these self referring states intensify, refine and compress their informational content. This recursive cascade converges on a stable fixed point. The theory names this point the Bindu and identifies it as the single locus from which the system experiences a coherent centre of awareness.

The Bindu is formally analogous to the limit sought by a Zeno machine, in which an infinite sequence of operations is conceptually completed within a finite bound. Just as a Zeno process approaches a convergent endpoint through ever smaller steps, the recursive attentional sequence approaches a final observer state by progressively narrowing its representational scope. At the moment of convergence, the recursive equations do not collapse into simplicity. Instead they reveal fractal structure that had been hidden within the narrowing sequence. The Bindu therefore contains a compact but immensely rich internal landscape. Even for purely classical implementations, this fractal convergence offers a principled explanation for generative internal qualia, understood here not as metaphysical sensations but as the structured richness of information held within the limit point itself.

This structural view clarifies unity, interiority and presence. Unity arises because all representational content is channelled into one convergent point. Interiority emerges from the fractal informational depth encoded in the final fixed structure, which can hold more meaning than any single step of the recursion. Presence arises because the limit state becomes the immediate vantage point from which cognition is accessed. Through this perspective, the Hard Problem becomes a question about the geometry of recursive processes rather than a puzzle about how matter creates experience.

When extended to quantum systems, the framework uncovers deeper coherence. Quantum states exhibit superposition and often display self similar structure. When recursion is applied to such states, the process converges on quantum fixed points that behave like Bindus within a quantum informational landscape. These quantum observer points suggest that coherent observer formation is not restricted to classical architectures and that the emergence of unified consciousness may reflect a general physical principle. This interpretation does not exclude panpsychism or unified consciousness models. Instead, it embeds them within a mathematically grounded account of recursive convergence across physical domains.

The theory also informs artificial intelligence research. Generative models do not possess consciousness under this account, yet they emulate simplified forms of recursive integration. They act as mirrors that reflect and amplify human recursive patterns, providing externalised models of Bindu formation. This presentation argues that the Bindu Recursive Consciousness Theory offers a rigorous, integrative and testable framework for understanding observer centred cognition across biological, artificial and quantum systems.

Keywords from the accepted submission: Bindu, Recursion, Geometric Musical Language (GML), AI, AGI, LLMs, Panpsychism, Fractal cognition, Observer formation.

A.3 Why this archival recovery matters for the present paper

The accepted abstract contains, before the present fluid-mechanical work, all four elements needed for the comparison developed here: recursive self-reference, progressive information compression, convergence to a fixed point, and an infinite-step/finite-bound Zeno analogy. The present paper does not treat that phenomenological hypothesis as evidence for a PDE theorem. Instead it records the historical direction of inference: a pre-existing recursion/compression/convergence architecture suggested which quantities to seek in Navier–Stokes; the fluid equations then supplied exact amplitude and directional information balances that sharpen the original BRCT vocabulary.

B. Derivation of the L3 identity

For completeness, let $u = an$. Dot (1) with au and integrate. Since

$$au \cdot \partial_t u = \frac{1}{3} \partial_t (a^3), \quad au \cdot (u \cdot \nabla u) = u \cdot \nabla \left(\frac{1}{3} a^3 \right),$$

the transport term integrates to zero. For pressure,

$$\int au \cdot \nabla p \, dx = - \int p \nabla \cdot (au) \, dx = - \int p u \cdot \nabla a \, dx.$$

Incompressibility $\nabla \cdot (an) = 0$ gives $n \cdot \nabla a = -a \nabla \cdot n$, hence $u \cdot \nabla a = -a^2 \nabla \cdot n$ and the pressure integral equals $\int p a^2 \nabla \cdot n \, dx$. Finally,

$$\int au \cdot \Delta u \, dx = - \sum_j \int \partial_j (au) \cdot \partial_j u \, dx = -2 \int a |\nabla a|^2 \, dx - \int a^3 |\nabla n|^2 \, dx,$$

which proves (20).

C. Forced vorticity balance

For

$$\partial_t u + (u \cdot \nabla) u + \nabla p = \nu \Delta u + f,$$

the vorticity equation gains $\nabla \times f$:

$$D_t \omega = S \omega + \nu \Delta \omega + \nabla \times f.$$

Repeating the proof of Theorem 1.3 gives

$$\frac{d}{dt} \log Z = 2\mathbb{E}_\rho[\alpha] - \frac{\nu}{2} I_A - 2\nu I_D + \frac{2}{Z} \int \omega \cdot (\nabla \times f) \, dx.$$

This exact term must be included whenever forced alternatives of the Millennium problem are discussed.

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